

Carcinoembryonic Antigen (CEA)

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

This assay is intended for the in vitro quantitative determination of carcinoembryonic antigen in human serum and plasma.¹ The Elecsys CEA assay is further indicated for serial measurement of CEA to aid in the management of cancer patients. The main indication for CEA determinations is the follow-up and therapy-management of colorectal carcinoma.

Special Instructions

Values obtained with different assay methodologies should not be used interchangeably in serial testing. It is recommended that only one assay method be used consistently to monitor each patient's course of therapy. This procedure does not provide serial monitoring; it is intended for one-time use only. If serial monitoring is required, please order the serial monitoring test (480095 (</tests/480095/carcinoembryonic-antigen-cea-serial-monitor>)).

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

CEA determinations are not recommended for cancer screening in the general population. CEA concentrations within the normal range do not exclude the possible presence of a malignant disease.

The measured CEA value of a patient's sample can vary depending on the test procedure used. CEA values determined on patient samples by different test procedures cannot be directly compared with one another and could be the cause of erroneous medical interpretations.

As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or have received them for diagnostic purposes.¹

In rare cases, interference due to extremely high titers of antibody to streptavidin and ruthenium can occur. The test contains additives which minimize these effects.¹

For diagnostic purposes, the results should always be assessed in conjunction with the patient's medical history, clinical examination, and other findings.

Custom Additional Information

CEA is a monomeric glycoprotein (molecular weight approximately 180,000 daltons) with a variable carbohydrate component of approximately 45% to 60%.^{1,2} CEA, like AFP, belongs to the group of carcino-fetal antigens produced during the embryonic and fetal period. CEA is mainly found in the fetal gastrointestinal tract and in fetal serum. It also occurs in slight quantities in intestinal, pancreatic, and hepatic tissues of healthy adults. The formation of CEA is repressed after birth and, accordingly, serum CEA values are hardly measurable in healthy adults.

High CEA concentrations are frequently found in cases of colorectal adenocarcinoma.^{3,4} Slight to moderate CEA elevations (rarely >10 ng/mL) occur in 20% to 50% of benign diseases of the intestine, pancreas, liver, and lungs (eg, liver cirrhosis, chronic hepatitis, pancreatitis, ulcerative colitis, Crohn's disease, emphysema).^{3,5} Smokers also have elevated CEA values. The reactive epitopes of CEA have been characterized, and the available monoclonal antibodies classified, into six epitope groups.⁶⁻⁸ The antibodies used in the Elecsys CEA assay react with epitopes two and five. The antibodies react with CEA and (as with almost all CEA methods) with the meconium antigen (NCA2).⁸ Cross-reactivity with NCA1 is 0.7%.

Specimen Requirements

Specimen

Serum

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Gel-barrier tube (preferred) or red-top tube

Collection Instructions

If a red-top tube is used, transfer separated serum to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	7 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

Range: 0–4.7 ng/mL; nonsmoker: <3.9 ng/mL; smoker: <5.6 ng/mL

Storage Instructions

Room temperature

Causes for Rejection

Citrate plasma specimen; improper labeling

References

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Footnotes

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